



Republic of Rwanda
Ministry of Education



ROOF CONSTRUCTION

BDCRC501

Construct Roof Structure

Competence

RQF Level: 5

Learning Hours



Credits: 8

Sector: Construction and Building Services

Trade: Building Construction

Module Type: Specific

Curriculum: CBSBDC5001-TVET CERTIFICATE 5 IN BUILDING CONSTRUCTION

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Purpose statement	This module describes skills, knowledge and attitudes required to construct a roof structure. It is meant for learners who have successfully completed level IV in building construction and pursuing level V in building construction. At the end of this module students will be able to Perform roof preliminary works, set out the roof, fix roof and apply roof coverings with minimum supervision.					
Leaning assumed to be in place	Scaffolding operations, Basic reinforced concrete design, ARCHICAD software.					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	• Group project and presentation	20%				
	• Individual project /Work	50%				
			Summative Assessment			50%

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
Perform roof preliminary works	1.1. Roof plan is correctly interpreted based on drawing details.
	1.2. Safety precautions are effectively respected/applied according to the working condition.
	1.3. Materials, tools and equipment are properly prepared based on project specifications.
Set out roof	2.1. Working place is adequately prepared according to the site condition
	2.2. Method of setting out is appropriately applied according to the type of roof
	2.3. Measurements of roof structure are properly checked referring to the roof plan.
	2.4. Reference line is accurately fixed according to the type of roof

	2.5. Roof slope is properly regulated according to the roof plan
	2.6. Size of gutters are properly determined based on roof catchment area
	2.7. Gutters and valleys are properly set according to drawings and specification
Fix roof structure	3.1. Shape of trusses is properly identified according to drawings and specifications.
	3.2. Truss members are correctly assembled according to drawings and specifications.
	3.3. Roof members are properly fixed according to drawings and specifications
	3.4. Protective materials are properly applied according to the type of roof material and specifications
	3.5. Stability of roof structure is effectively ensured according to the techniques of fixing
Apply roof coverings	4.1. Status of materials is adequately verified according to the specifications.
	4.2. Drainage gutters and valleys are appropriately installed according to roof drawings.
	4.3. Insulation materials are installed according to roof drawings.
	4.4. Roof coverings and ridge covers are properly fixed according to roof drawings.
	4.5. Protective/Sealant materials are properly applied based on project specification.
	4.6. Roof is appropriately checked in accordance with the roof drawing.

Intended Knowledge, Skills and Attitude

Knowledge	Skills	Attitude
✓ Describe the methods of removing hazard and risk ✓ Describe types of materials, tools and equipment required in construction ✓ Identify shape of trusses ✓ Identify parts of roof ✓ Describe the process of fixing roof and roof coverings ✓ Describe insulation materials for roof covering ✓ Identify methods of setting out roof	✓ Perform cutting methods ✓ Interpret construction drawings ✓ Perform assembling techniques ✓ Carry out finishing activities ✓ Set out roofs ✓ Apply roof coverings ✓ Manage the wastes ✓ Prepare technical report of the work done	✓ Use materials efficiently and effectively ✓ Demonstrate punctuality during the implementation of design project ✓ Demonstrate resourcefulness in the new design trends ✓ Demonstrate effective communication skills to stakeholders. ✓ Manage time effectively to meet project timelines ✓ Having teamwork and collaboration spirit ✓ Demonstrate ethical practices at the workplace ✓ Demonstrate willingness to adapt to changes in roof works ✓ Practice compliance with health, safety, social and environmental regulations ✓ Listen to clients' needs open-mindedly ✓ Pay attention to details while making a roof

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Perform roof preliminary works 2. Set out roof structures 3. Fix roof structure 4. Apply roof coverings
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Learning outcome 1: Perform roof preliminary works	Learning hours: 10 hours
Indicative content	
• Interpretation of roof plan ✓ Drawing dimensions ✓ Drawing symbols and signs ✓ Drawing scale ✓ Drawing plans ✓ Drawing sections and views • Applying safety precautions in working area ✓ Use PPE's ✓ Set up Working area safety measures ✚ Sign posts ✚ Hazards and risks management • Selection of materials, tools and equipment ✓ Classification of tools used for roof construction ✚ Holding tools ✚ Setting out tools ✚ Cutting and shaving tools ✚ Boring tools ✚ Fixing tools ✚ Calibrated measuring tools ✚ Power tools ✚ Levelling tools ✚ Miscellaneous tools ✓ Classification of equipment ✚ Portable machines ✚ Heavy duty machine ✓ Identification of materials used in roof construction ✚ Types of roof materials ✚ Factors to consider when choosing roof materials ✚ Types of roof coverings materials ✚ Quantification of roof materials	

Resources required for the indicative content

Equipment	PPE, Circular saw , Band saw, Surface planer, Thicknesses planner, Belt sander ,Nail gun, Ladder, Drilling machine, Dumpy level, Wheel barrow, Grinder, Welding machine, scaffolding, laser meter, theodolite, roofing shear.
Materials	Iron sheet, Roof tiles, Wooden board, Nails, Screws, Bolt and nuts, Galvanized wire, Glue, Cutting disc, Grinding disc, Varnish, Lacquer, water tank,Thinner, Cement, aggregates, nails, Silicone, wooden sealant materials
Tools	Club hammer, claw hammer, Plumb bob, Building line, Safety belt, Measuring tape, Spirit level, Hand saws, Machete, Folding rulers, Screw driver kit, Spanners, Crow bar, Hand planes, Squares, Brush, Chisels, Files, hose pipe, Rasps, Marking gauge
Facilitation techniques	Group discussion, scenario, simulation, individual activity - The trainer prepares a roof plan and he/she asks students to identify elements to inspect while interpreting the roof plan, - The trainer sets up a scenario related to activities in roof construction and he /she asks students to identify potential hazards associated with the roof activity, - The trainer provides a list of tools and equipment used in roof construction and he /she asks students to categorize them accordingly - The trainer gives a roof plan including all data and he /she asks students to quantify the required roof materials
Formative assessment methods	- oral assessment - Written assessment - Performance assessment

Indicative content

- **Preparation of working place**
 - ✓ Conduction of site visit
 - ✚ Identification of type of buildings
 - ✚ Identification of types of roof
 - ✓ Cleaning the working place
 - ✓ Checking the top of the wall for accommodating the wall plate
 - ✓ Recording measurements of the trusses
- **Applying methods of setting out of roof frame**
 - ✓ Direct setting out/lay out
 - ✓ Steel square method
 - ✓ 3,4,5 methods
 - ✓ Levelling instruments
- **Checking the measurements of roof structure**
 - ✓ Identification of roof structure components
 - ✚ Ridged board
 - ✚ Common rafter
 - ✚ Purlin
 - ✚ Valley rafter
 - ✚ Cripple rafter
 - ✚ Verge
 - ✚ Hip rafter
 - ✚ Eaves
 - ✚ Crown rafter
 - ✚ Jack rafter
 - ✚ Wall plate
 - ✓ Determine the roof height
 - ✓ Determine roof slope
 - ✓ Determine the roof area
 - ✓ Measure the distance the roof extends beyond the walls
- **Fixing reference line of roof**
 - ✓ Methods of fixing reference line of roof
 - ✓ Methods of fixing pegs for trusses
 - ✓ Steps of fixing reference line for roof
- **Regulating the roof slope**
 - ✓ Install roof rafter and trusses
 - ✓ Adjust roof slope
- **Determination of sizes of gutters**
 - ✓ Check for the gutter size
 - ✓ Types of gutters
 - ✓ Determination of gutter size

-  Catchment area determination
-  Determination of rainfall intensity
-  Determine the gutter size

- **Setting gutters and valleys**

- ✓ Set out roof valleys
- ✓ Set out roof gutters

Resources required for the indicative content

Equipment	PPE, Circular saw , Band saw, Surface planer, Thicknesses planner, Belt sander ,Nail gun, Ladder, Drilling machine, Dumpy level, Wheel barrow, Grinder, Welding machine, scaffolding, laser meter, theodolite, roofing shear.
Materials	Iron sheet, Roof tiles, Wooden board, Nails, Screws, Bolt and nuts, Galvanized wire, Glue, Cutting disc, Grinding disc, Varnish, Lacquer, Thinner, Cement, aggregates, nails, Silicone, wooden sealant materials
Tools	Club hammer, claw hammer, Plumb bob, Building line, Safety belt, Measuring tape, Spirit level, Hand saws, Machete, Folding rulers, Screw driver kit, Spanners, Crow bar, Hand planes, Squares, Brush, Chisels, Files, hose pipe, Rasps, Marking gauge.
Facilitation techniques	Group discussion, scenario, simulation, individual activity - The trainer shows a video demonstrating the preparation of a work area before starting to construct the roof and he/she asks students to summarize the main activities involved in this preparation - The trainer presents a video demonstrating the process of setting out a roof and divides the students into manageable groups, prompting them to discuss the steps involved in setting out the roof. - Trainer provides data of roof structure and asks trainees to determine the size of gutters. - The trainer prepares the steps for installing gutters and he/she asks the students to carry out these activities - The trainer gives students data on roof structure components and requests them to calculate the roof area, slope, and height - Trainer and students perform the activity of setting out the reference line and the trainer asks trainees to set the reference line themselves. - Trainer and students regulate the roof slope and trainer asks trainees to regulate the roof slope. - Trainer prepares the field visit to the ongoing project site.

Formative assessment methods	• Oral assessment • Written assessment • Performance assessment • Field visit report
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Learning outcome 3: Fix roof structure	Learning hours: 25 hours
Indicative content	
• Assembling truss members ✓ Identification of Shape of trusses ✚ King post truss ✚ Queen post truss ✚ Composite truss ✚ Steel truss ✚ Mansard truss ✓ Assembling technics ✓ Assembling truss members ✚ Planning ✚ Measuring ✚ Marking ✚ Cutting ✚ Jointing • Fixing roof members ✓ Lifting trusses to their positions ✚ Methods of lifting ✚ Method of equilibrium ✓ Fixing purlins and braces ✓ Keeping trusses in the plumb position ✚ Use of a plumb bob ✚ Vertical/horizontal positions • Application of preservative materials ✓ Types of materials ✚ Oil materials ✚ Chemicals ✚ Solvents ✓ Method of applying preservative materials ✚ Brushing ✚ Spray ✚ Gun ✚ Rolling • Ensuring stability of roof structures ✓ Check the structural supports of roof	

 Rafters  Trusses  Beams ✓ Incorporate bracings and reinforcements	
Resources required for the indicative content	
Equipment	PPE, Circular saw , Band saw, Surface planer, Thicknesses planner, Belt sander ,Nail gun, Ladder, Drilling machine, Levelling tools, Grinder, Welding machine, scaffolding
Materials	Iron sheet, Wooden board, Nails, Screws, Bolt and nuts, Galvanized wire, Warning tapes, Glue, Cutting disc, Grinding disc, Varnish, Lacquer, Thinner, Cement, Water, Silicone, Paint, Notebook
Tools	Club hammer, claw hammer, Plumb bob , Building line, Carpenter’s bags, Safety belt, Measuring tape, Spirit level, Hand saws , Machete , Folding rulers ,Screw driver kit, Spanners, Hand planes , Squares , Brush , Chisels , Divider, Files, Rasps, Marking gauge
Facilitation techniques	Group discussion, scenario, simulation, individual activity - Trainer and students perform the activity of assembling truss members and he/she asks students to perform assembling truss members themselves individually or in groups. - The trainer and trainees perform fixing roof members and the trainer asks students to fix roof members themselves individually or in groups. - The trainer provides a video of applying roof sealant materials and he/she asks students to identify the sealant material used. - Trainer and students to perform application of roof sealant material and trainer asks students to apply sealant materials on the roof themselves. - Trainer prepares the field visit to the ongoing project site.
Formative assessment methods	- Oral assessment - Written assessment - Performance assessment - Field visit report

Learning outcome 4: Apply roof coverings		Learning hours: 25 hours
Indicative content		
• Installing Drainage gutters and valleys. ✓ Identification of Methods gutter fixing ✓ installing gutters and valley ✓ Inspect the slope and check drainage flow • Installing roof Insulation materials. ✓ Identification of Types of roof insulation materials ✓ Applying roof insulation materials ✚ Warm flat roofing ✚ Cold flat roofing • Fixing Roof coverings and ridge covers. ✓ Make Visual Inspection of roof coverings materials ✚ Defects of roof covering materials ✚ Size of roof covering materials ✓ Laying techniques and principles for roof coverings ✓ Fixing roof coverings techniques ✚ Using nails ✚ Screws ✚ Bolt and nuts • Applying Protective/Sealant materials. ✓ Identification Types of materials ✓ Applying wood preservatives • Checking Roof ✓ Identification of tools and materials used in roof checking ✓ Checking roof structure ✚ Inspect the inside of the roof. ✚ Check your ventilation. ✚ Look for loose and damaged roof material. ✚ Check the flashing. ✚ Unblock gutters. ✚ Check for moss growth		
Resources required for the indicative content		
Equipment	• PPE, Circular saw , Band saw, Surface planer, Thicknesses planner, Belt sander ,Nail gun, Ladder, Drilling machine, Levelling tools, Grinder, Welding machine, scaffolding	
Materials	• Iron sheet, Wooden board, Nails, Screws, Bolt and nuts, Galvanized wire, Warning tapes, Glue, Cutting disc, Grinding disc, Varnish, Lacquer, Thinner, Cement, Water, Silicone, Paint, Notebook	

Tools	• Club hammer, claw hammer, Plumb bob , Building line, Carpenter's bags, Safety belt, Measuring tape, Spirit level, Hand saws , Machete , Folding rulers ,Screw driver kit, Spanners, Hand planes , Squares , Brush , Chisels , Divider, Files, Rasps, Marking gauge
Facilitation techniques	Group discussion, scenario, simulation, individual activity • Trainer provides images of roof coverings and asks learners to name them. • Trainer brings damaged roof covering material and asks trainees to identify its defects. • Trainer displays a video of applying roof coverings and in manageable groups, he/she asks the students to discuss on steps involved in applying roof coverings • Trainer and trainees perform installing roof covering to the roof structure and trainer ask trainees to install roof covering themselves under supervision. • Trainer display video of checking roof and ask trainees to list down element checked • Trainer prepare the field visit to the ongoing project site.
Formative assessment methods	• Oral assessment • Written assessment • Performance assessment • Field visit report

Integrated/Summative assessment

Integrated situation

XL contractors located at KAMONYI District wants to construct gable roof based on an issue of the lack of timber in region especially in Ntarama Model village, but does not have qualified workers to construct that roof. In tendering process Magic Hands Company (MHC) Ltd won the tender for constructing accordingly. As the roof maker hired to construct the foresaid roof, you are requested to calculate required materials and construct gable roof of one chimney with the following information: roof will have 15 m x10m, the king post is of 3.5m and overhang 1.20m; the covering materials are iron sheets, the timber to be used for trusses is in eucalyptus. The following tasks will be done in group of six (6) trainees.

- ✓ Perform setting out of the roof by using square method.
- ✓ Prepare materials to the required sizes
- ✓ Assemble trusses in accordance with the standard
- ✓ Join and secure all trusses together
- ✓ Perform roof finishing

But before you effectively start the real work, you are asked to demonstrate your competences by making a small roof having 3m x2m, the riser of 3.5m and overhang 1.20m, same covering materials and same timber for trusses. All materials, tools and equipment are found in Magic Hands Company (MHC) Ltd

Drawing is also available.

The proofing work should be done within 10 hours.

Resources

Tools	Club hammer, claw hammer, Plumb bob , Building line, Carpenter's bags, Safety belt, Measuring tape, Spirit level, Hand saws , Machete , Folding rulers ,Screw driver kit, Spanners, Hand planes , Squares , Brush , Chisels , Divider, Files, Rasps, Marking gauge
Equipment	PPE, Circular saw , Band saw, Surface planer, Thicknesses planner, Belt sander ,Nail gun, Ladder, Drilling machine, Levelling tools, Grinder, Welding machine, scaffolding
Materials/ Consumables	Iron sheet, Wooden board, Nails, Screws, Bolt and nuts, Galvanized wire, Warning tapes, Glue, Cutting disc, Grinding disc, Varnish, Lacquer, Thinner, Cement, Water, Silicone, Paint, Notebook

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Learning outcome 1: Perform roof preliminary works	1.1. Roof plan is correctly interpreted based on drawing details	Roof dimensions are interpreted			3
		Roof symbols and sign are interpreted			3
	1.2. Safety precautions are effectively respected/applied according to the working condition.	PPE are worn			3
		Sign posts are installed			3
	1.3. Materials, tools and equipment are properly prepared based on project specifications.	Materials are selected			3
		Tools are selected			3
		Equipment are selected			3
Learning outcome 2: Set out roof	2.1. Working place is adequately prepared according to the site condition	Working place is cleaned			4
		The top of the wall is checked			4
		Measurements of the trusses are taken			4
		Square method is used			4

	2.2. Method of setting out is appropriately identified according to the type of roof	Steps of fixing reference line are respected			4
		Reference line is fixed			4
		Pegs for trusses are fixed			4
	2.3. Measurements of roof structure are properly checked referring to the roof plan.	Measurements of roof structures are taken			4
	2.4. Roof slope is properly regulated according to the roof plan	Slopes are calculated			4
	2.5. Size of gutters and valleys are properly determined based on roof catchment area	Sizes of gutters are calculated			5
Learning outcome 3: Fix roof structure	2.7. Gutters and valleys are properly set according to drawings and specification	Gutters and valleys are set			4
	3.1. Shape of trusses is properly identified according to drawings and specifications.	Shape of trusses is respected			5
	3.2. Truss members are correctly assembled according to drawings and specifications.	Truss members are assembled			5
		Truss members are stable			5
	3.3. Protective materials are properly applied according to the type of roof material and specifications	Protective materials are applied			4
	3.4. Stability of roof structure is effectively ensured according to the techniques of fixing	Roof structure is stable			4
Learning outcome 4: Apply roof coverings	4.1. Status of materials is adequately verified according to the specifications.	Status of roof materials are verified			4
	4.2. Roof coverings and ridge covers are properly fixed according to roof drawings.	Roof coverings and ridge covers are fixed			4

	4.3. Roof is appropriately checked in accordance with the roof drawing.	Roof structure is checked			4
Total marks		100			
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

References

1. Carpentry and joinery (Second Edition, volume3) by Brian Porter & Christopher Tooke 2.
2. BS 476-3: 2003, Fire tests on building materials and structures. Classification and methods of test for external fire exposure to roofs.
3. BS 8414-1: 2002, Fire Performance of external Cladding systems.
4. BS 7956: 2000, Specification for primers for woodwork.
5. BS 1282: 1999, Wood Preservatives. Guidance on choice use and application.
6. BS 4261: 1999, Wood Preservation, Vocabulary.
7. BS 5589: 1989, Code of practice for wood preservation.
8. Building regulations, Approved Document A, Structure: 2000.
9. BS 5395: Part 1: 2000, Stairs, ladders and walkways. Code of practice for the design, construction and maintenance if straight stairs and winder.
10. BS EN 923, 2005 Adhesives. Terms and definitions.
11. BS EN 1186-2, 1988, Timber for, and workmanship in Joinery. Specification for workmanship.
12. HSE (Health & Safety Executive) Manual handling leaflets.
13. Health and Safety at Work Act (HSAWA) 1974.